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CH.SC.U4CSE24207

WEEK 4

BFS Traversal:

Code:

```
#include <stdio.h>
#define MAX 100

int adj[MAX][MAX];
int visited[MAX];
int queue[MAX];
int front = 0, rear = 0;
int n;

void BFS(int start) {
    visited[start] = 1;
    queue[rear++] = start;
    while (front != rear) {
        int v = queue[front++];
        printf("%d ", v);
        for (int i = 0; i < n; i++) {
            if (adj[v][i] == 1 && !visited[i]) {
                visited[i] = 1;
                queue[rear++] = i;
            }
        }
    }
}
```

```

    }

    printf("\nBFS Traversal starting from 0:\n");
    BFS(0);

    return 0;
}

```

Output:

```

root@amma29:/home/amma/Downloads# gcc -o bfs bfs.c
root@amma29:/home/amma/Downloads# ./bfs
Enter number of vertices: 5
Enter number of edges: 4
Enter edges (u v):
0 1
0 2
1 3
1 4

BFS Traversal starting from 0:
0 1 2 3 4 root@amma29:/home/amma/Downloads#

```

DFS Traversal:

Code:

```
#include <stdio.h>
#define MAX 100

int adj[MAX][MAX];
int visited[MAX];
int n;

void DFS(int v) {
    visited[v] = 1;
    printf("%d ", v);
    for (int i = 0; i < n; i++) {
        if (adj[v][i] == 1 && !visited[i]) {
            DFS(i);
        }
    }
}

int main() {
    int edges, u, v;

    printf("Enter number of vertices: ");
    scanf("%d", &n);

    printf("Enter number of edges: ");
    scanf("%d", &edges);

    for (int i = 0; i < n; i++)
        for (int j = 0; j < n; j++)
            adj[i][j] = 0;

    printf("Enter edges (u v):\n");
    for (int i = 0; i < edges; i++) {
        scanf("%d %d", &u, &v);
        adj[u][v] = adj[v][u] = 1;
    }

    printf("\nDFS Traversal starting from 0:\n");
    DFS(0);

    return 0;
}
```

Output:

```
0 root@amma29:/home/amma/Downloads# gcc -o dfs dfs.c
root@amma29:/home/amma/Downloads# ./dfs
Enter number of vertices: 5
Enter number of edges: 4
Enter edges (u v):
0 1
0 2
1 3
1 4

DFS Traversal starting from 0:
0 1 3 4 2 root@amma29:/home/amma/Downloads#
```